Abstract algebra – 2

Finite rings and division rings

Homework

13 сентября 2026 года

Theoretical minimum. A ring is an abelian group with an associative multiplication distributing over addition; a unit is an invertible element, a zero divisor kills a non-zero element, a is nilpotent if $a^k = 0$ and idempotent if $a^2 = a$; the characteristic is the least $n > 0$ with $n \cdot 1 = 0$. In a *finite* ring the maps $x \mapsto ax$ and $x \mapsto xa$ are injective exactly when they are surjective, so every element is either a unit or a two-sided zero divisor, and a finite ring without zero divisors is a division ring; by Wedderburn's theorem a finite division ring is commutative, hence a field. If $\operatorname{char} A = p$ then $(A, +)$ is an $\mathbb{F}_p$-vector space and $|A| = p^k$, and in the commutative case $x \mapsto x^p$ is an endomorphism. A Boolean ring ($x^2 = x$) is commutative of characteristic 2; more generally an identity $x^n = x$ forces commutativity (Jacobson). If a is nilpotent then $1 - a$ is a unit; $1 - ab$ is a unit if and only if $1 - ba$ is, by the identity $(1 - ba)^{-1} = 1 + b(1 - ab)^{-1}a$.

---

1. Prove that in an arbitrary ring $1 - ab$ is invertible if and only if $1 - ba$ is, and verify the explicit formula for the inverse. Deduce that ab is nilpotent if and only if ba is, and give an example showing that the sum of two nilpotent elements of a non-commutative ring need not be nilpotent.

2. Let R be a finite commutative ring. Prove that (a) the set of nilpotent elements is closed under addition and multiplication; (b) the number of nilpotent elements divides the number of zero divisors (here 0 is also counted as a zero divisor). (Румыния, 2009)

3. Let R be a ring of characteristic zero (meaning that $nx = 0$ with $n \in \mathbb{Z} \setminus \{0\}$ implies $x = 0$), not necessarily commutative, and let e, f, g be idempotents with $e + f + g = 0$. Show that $e = f = g = 0$. (IMC)

4. Let R be a unital ring, $n \geqslant 2$, and let $x \in R$ satisfy $x^n = x$. Prove that for every $m \geqslant 1$ there is a unit $u_m \in R^*$ with $x^m = u_m x$.
(IMC 2025, shortlist)

5. Determine all non-trivial finite rings with a unit element in which the sum of all elements is invertible. (Румыния, 2021)

6. Determine all rings $(A, +, \cdot)$ such that $a^2 \in \{0, 1\}$ for every $a \in A$.
(Румыния, 2022)

7. Let $(K, +, \cdot)$ be a division ring in which $x^2 y = yx^2$ for all $x, y \in K$. Prove that K is commutative. (Румыния, 2024)

8. Let A be a ring with $2n+1$ elements and let $M = \{k \in \mathbb{Z} : k \geqslant 2,\ x^k = x \text{ for all } x \in A\}$. Prove that A is a field if and only if M is non-empty and its smallest element equals $2n + 1$. (Румыния, 2005)

9. Prove that the multiplicative group of a finite field with q elements is cyclic. Deduce that for every divisor d of $q - 1$ the equation $x^d = 1$ has exactly d solutions, and that the sum of all elements of a finite field is zero except for the field with two elements. (The equivalence "$1+1 = 0$ if and only if $f(X^2)$ is reducible for all $\deg f > 1$" is Polynomials – 6, homework 4.)

10. Let A be a ring in which every element is a product of two idempotent elements. Show that (a) 1 is the only unit of A; (b) A is Boolean.
(Румыния, 2015)